## Part 2: Technical Document

### 1. Entity Relationship Diagram (ERD) Structure

The data model uses an optimized Star Schema cantered around a core fact table and connected lookup dimensions:

• Fact Table: Hospital (Primary Key: Facility ID)

• Dimension Tables (1:N Relationships via Direct Keys):

- State_Lookup connected via the State column.
- County_Lookup connected via the County column.
- HospitalType_Lookup connected via the Hospital Type column.
- Ownership_Lookup connected via the Hospital Ownership column.
- PerformanceDomain_Bridge (Unpivoted table for domain-level measure distributions).

### 2. Power Query Data Transformation Steps

a. Source Ingestion: Connected directly to Hospital_General_Information.csv with source text encoding verified.
b. Column Standardization: Cleaned and sanitized all 38 original column headers, removing spaces, slashes, and special characters to ensure robust syntax.
c. Data Cleaning & Type Casting:
   - Replaced non-numeric placeholder values in Hospital overall rating with Null blanks and cast the column as a whole number integer.
   - Standardized the EmergencyServices field into a uniform, logical Yes/No binary string layout.
   - Applied Text.Trim globally across all text-based fields to remove erroneous leading or trailing white spaces.

### 3. DAX Measures

**Basic KPIs:**

| KPIs | Functional Description / Scope | Und`erlying DAX Formula / Model Logic |
|---|---|---|
| **Total Hospitals** | Displays the total volume of active hospital records within the currently selected filter context. | `MEASURE _Measures [Total Hospitals]` *(or base count variant* `[Rated Hospitals]`*)* |
| **% High-Performing** | Calculates the current percentage of facilities scoring an overall quality rating equal to or above the threshold criteria. | `MEASURE _Measures [% High-Performing] = DIVIDE ([High-Performing Hospitals], [Rated Hospitals])` |
| **Avg % Better (Safety & READM)** | Computes the composite performance forecast score by averaging the projected clinical improvements across safety and return-visit tracking domains. | `MEASURE _Measures [Avg % Better (Safety & READM)] = AVERAGEX ({[Safety Projected % Better], [READM Projected % Better]}, [Value])` |
| **% Birthing-Friendly** | Tracks the dynamic proportion of hospital facilities categorized as birthing-friendly out of the filtered dataset. | `MEASURE _Measures [% Birthing-Friendly]` |

## Domain metrics for each performance domain.

| Performance Domain | Functional Scope / Description | Implemented Measure Name | Exact DAX Formula |
|---|---|---|---|
| **Mortality Domain (MORT)** | Tracks forward-looking volume and percentage distribution shifts concerning patient mortality rates. | MORT Projected Worse | MEASURE _Measures [MORT Projected Worse] = [MORT Worse] * (1 - [Measure Improvement Value]) |
| **Mortality Domain (MORT)** | Tracks forward-looking volume and percentage distribution shifts concerning patient mortality rates. | MORT Projected Better | MEASURE _Measures [MORT Projected Better] = [MORT Better] + [MORT Worse] * [Measure Improvement Value] * [Shift Share to Better] |
| **Mortality Domain (MORT)** | Tracks forward-looking volume and percentage distribution shifts concerning patient mortality rates. | MORT Projected No Different | MEASURE _Measures [MORT Projected No Different] = [MORT No Different] + [MORT Worse] * [Measure Improvement Value] * (1 - [Shift Share to Better]) |
| **Mortality Domain (MORT)** | Tracks forward-looking volume and percentage distribution shifts concerning patient mortality rates. | MORT Projected % Better | MEASURE _Measures [MORT Projected % Better] = DIVIDE ([MORT Projected Better], [MORT Measures Reported]) |
| **Mortality Domain (MORT)** | Tracks forward-looking volume and percentage distribution shifts concerning patient mortality rates. | MORT Projected % No Different | MEASURE _Measures [MORT Projected % No Different] = DIVIDE ([MORT Projected No Different], [MORT Measures Reported]) |
| **Mortality Domain (MORT)** | Tracks forward-looking volume and percentage distribution shifts concerning patient mortality rates. | MORT Projected % Worse | MEASURE _Measures [MORT Projected % Worse] = DIVIDE ([MORT Projected Worse], [MORT Measures Reported]) |
| **Safety of Care Domain (Safety)** | Tracks complication modeling projections and dynamic hospital clinical safety event metrics. | Safety Projected Worse | MEASURE _Measures [Safety Projected Worse] = [Safety Worse] * (1 - [Measure Improvement Value]) |
| **Safety of Care Domain (Safety)** | Tracks complication modeling projections and dynamic hospital clinical safety event metrics. | Safety Projected Better | MEASURE _Measures [Safety Projected Better] = [Safety Better] + [Safety Worse] * [Measure Improvement Value] * [Shift Share to Better] |
| **Safety of Care Domain (Safety)** | Tracks complication modeling projections and dynamic hospital clinical safety event metrics. | Safety Projected No Different | MEASURE _Measures [Safety Projected No Different] = [Safety No Different] + [Safety Worse] * [Measure Improvement Value] * (1 - [Shift Share to Better]) |
| **Safety of Care Domain (Safety)** | Tracks complication modeling projections and dynamic hospital clinical safety event metrics. | Safety Projected % Better | MEASURE _Measures [Safety Projected % Better] = DIVIDE ([Safety Projected Better], [Safety Measures Reported]) |
| **Safety of Care Domain (Safety)** | Tracks complication modeling projections and dynamic hospital clinical safety event metrics. | Safety Projected % No Different | MEASURE _Measures [Safety Projected % No Different] = DIVIDE ([Safety Projected No Different], [Safety Measures Reported]) |
| **Safety of Care Domain (Safety)** | Tracks complication modeling projections and dynamic hospital clinical safety event metrics. | Safety Projected % Worse | MEASURE _Measures [Safety Projected % Worse] = DIVIDE ([Safety Projected Worse], [Safety Measures Reported]) |
| **Readmission Domain (READM)** | Tracks projected percentage and volume frequency allocations relating to return visits and patient readmissions. | READM Projected Worse | MEASURE _Measures [READM Projected Worse] = [READM Worse] * (1 - [Measure Improvement Value]) |
| **Readmission Domain (READM)** | Tracks projected percentage and volume frequency allocations relating to return visits and patient readmissions. | READM Projected Better | MEASURE _Measures [READM Projected Better] = [READM Better] + [READM Worse] * [Measure Improvement Value] * [Shift Share to Better] |
| **Readmission Domain (READM)** | Tracks projected percentage and volume frequency allocations relating to return visits and patient readmissions. | READM Projected No Different | MEASURE _Measures [READM Projected No Different] = [READM No Different] + [READM Worse] * [Measure Improvement Value] * (1 - [Shift Share to Better]) |
| **Readmission Domain (READM)** | Tracks projected percentage and volume frequency allocations relating to return visits and patient readmissions. | READM Projected % Better | MEASURE _Measures [READM Projected % Better] = DIVIDE ([READM Projected Better], [READM Measures Reported]) |

| | | | |
|---|---|---|---|
| **Readmission Domain (READM)** | Tracks projected percentage and volume frequency allocations relating to return visits and patient readmissions. | READM Projected % No Different | MEASURE _Measures [READM Projected % No Different] = DIVIDE ([READM Projected No Different], [READM Measures Reported]) |
| **Readmission Domain (READM)** | Tracks projected percentage and volume frequency allocations relating to return visits and patient readmissions. | READM Projected % Worse | MEASURE _Measures [READM Projected % Worse] = DIVIDE ([READM Projected Worse], [READM Measures Reported]) |

### 4. Instructions for Adding New Performance Domains

To scale the architecture for additional quality metrics (e.g., Timeliness of Care - TE):

I. Open the Power Query Editor, navigate to the raw source data table, and identify the measure count columns for the target domain.
II. Unpivot the new domain's "Better", "No Different", and "Worse" count fields into the existing structured PerformanceDomain_Bridge table layout.
III. Click apply, then write three corresponding base measures in DAX to calculate percentages: PctTEBetter, PctTENoDifferent, and PctTEWorse.
IV. Incorporate the new domain series directly into the clustered column charts found on the Performance Measure Details and Detailed Hospital View pages.